

MOSTOFA HASIN MAHDI

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EDUCATION

Shanto-Mariam University of Creative Technology
Bachelor of Science in Computer Science and Engineering
CGPA: 3.96

Dhaka, Bangladesh
Nov 2026

PROJECTS

NAB Preg AI
Lead AI Engineer

FastAPI, Next.js, XGBoost, Scikit-Learn, MultiOutputClassifier, Mistral AI, Supabase

- Trained a **Multi-Output XGBoost Classifier** predicting 3 clinical targets (Risk, Nutrition, Anomaly) simultaneously in a single <50ms inference pass, injecting WHO-aligned **Clinical Boundary Anchors** to resolve majority-class bias.
- Developed an **OCR pipeline** integrating Mistral's **Pixtral-12B vision model** via a FastAPI proxy, parsing clinical prescriptions and lab reports into sanitized JSON payloads for frontend validation.
- Engineered a **5-step OCR failover** pipeline cycling through 4 backup Mistral API keys before routing to a local client-side Tesseract.js parser as a final fallback, guaranteeing **100% uptime** for report extractions.
- Implemented dual-layer validation using Zod (client) and Pydantic (server) to sanitize 10 distinct patient fields, preventing malformed payload insertion into the database.
- Demo:* [Live Link](#) | [GitHub Link](#)

PDF to Markdown Conversion Engine
CPU Bound CLI Tool

PyMuPDF, pdfplumber, Pandas, Tabulate, spaCy, PyTesseract

- Developed an open-source Python CLI tool leveraging a custom layout-detection algorithm with coordinate bounding boxes to accurately sort text, detect headings, and prevent overlapping content.
- Integrated **NLP-based text cleaning** (spaCy) to resolve sentence fragmentation, and implemented an **OCR fallback system** (PyTesseract) to process scanned pages as high-resolution images.
- Built an algorithmic table extraction tool that identifies complex grids and reconstructs them into aligned Markdown tables.
- Demo:* [GitHub Link](#) | [Release Link](#)

Bizit Business Suite
Multi-Tenant SaaS Platform

React, FastAPI, PostgreSQL, Neon DB, Render

- Isolated data across **5 test tenants** in a single PostgreSQL schema using Row-Level Security (RLS) to support concurrent access with **zero cross-tenant leaks**.
- Built a **3-tier RBAC system** (Admin/Manager/Viewer) and utilized Pydantic validation across 24 API endpoints to ensure malformed payloads never reach the database.
- Designed real-time finance dashboards that aggregate sales, costs, and profit in **under 500ms**, replacing the manual spreadsheet workflow which takes business owners 30 minutes per day.
- Demo:* username: hasin, password: hasin123 | [Live Link](#) | [GitHub Link](#)

TECHNICAL SKILLS

Languages:	Python, JavaScript, TypeScript, C++, SQL (PostgreSQL/MySQL)
Frameworks:	React, Next.js, FastAPI
AI/ML:	XGBoost, Pandas, Scikit-Learn, OCR (Mistral Pixtral, PyTesseract), NLP (spaCy)
Databases:	Supabase, PostgreSQL, Neon DB, MongoDB, MySQL
Tools:	Git, GitHub, Docker, SwaggerUI, Linux Shell Scripting

ACHIEVEMENTS

- Hackathons & Showcases:** Finalist at CloudcampBD Infinity AI BuildFest 2026, Champion of SMUCT Software Project Showcase 2026.
- Programming Contests:** Second Runner-up at SMUCT CSE FEST V2.0 Programming Contest, Participated in BUET & SUST IUPC Prelims, ICPC Prelim Regional Contests (2022–2023).